

Symbolic Regression Discovers the Radial Acceleration Relation: No Evidence for a MOND $\sqrt{g}$ Asymptote from Kinematic Data

ABSTRACT

We apply symbolic regression via PySR to 175 SPARC galaxies (3,391 points, $g_{\text{bar}} \in [10^{-13}, 10^{-8}] \text{ m/s}^2$) to discover the functional form of the radial acceleration relation (RAR) $g_{\text{obs}} = F(g_{\text{bar}})$ without theoretical priors. The data-driven search independently converges on the 2-parameter form $\log g_{\text{obs}} = a + b/\log g_{\text{bar}}$ (CPX5), with $a = -17.060 \pm 0.133$, $b = -72.71 \pm 1.38$ from bootstrap resampling. CPX5 outperforms all MOND interpolating functions on information criteria ($\Delta\text{AIC} = 88$ vs. best MOND IF, $\Delta\chi^2 \approx 960$). The form is stable across 3 independent SR seeds, robust to M/L assumptions (7–16% parameter variation vs. 580% for MOND a_0), and validated by blind recovery tests (2/3 MOND forms recovered within 4%). Joint fitting with weak-lensing data (Mistele+2024) extends the RAR by 2.5 dex (6.5 dex total), where CPX5 dominates with a PySR score of 3.85 (next best: 0.48). A direct test of the MOND $\sqrt{g_{\text{bar}}}$ asymptote yields $c = 0.10 \pm 0.15$ ($\Delta\chi^2 = 0.18$, $p = 0.67$), showing no evidence in SPARC data. External field effect tests find a correlation opposite to MOND’s prediction ($\rho = +0.106$). All models exhibit large intrinsic scatter ($\chi^2_{\text{red}} \approx 10$), reflecting genuine physical dispersion in the RAR. We conclude that CPX5 is the best empirical description of the RAR over the kinematic dynamic range, and that the MOND deep-MOND limit remains unconstrained by current rotation curve data.

1. INTRODUCTION

Galaxy rotation curves consistently show discrepancies between the gravitational acceleration inferred from baryonic matter and the observed centripetal acceleration (Rubin et al. (1978); Bosma (1981)). This discrepancy is typically described by either a dark matter (DM) halo or by modified Newtonian dynamics (MOND) (Milgrom (1983); McGaugh et al. (2016)). Both approaches have strengths: DM fits the cosmic microwave background while MOND predicts galaxy dynamics with one universal acceleration scale $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$.

McGaugh et al. (McGaugh et al. (2016)) showed that centripetal and baryonic accelerations correlate tightly across 153 SPARC galaxies, forming the radial acceleration relation (RAR). The functional form of this relation is itself the MOND interpolating function (IF) in log-log space:

$$g_{\text{obs}} = \frac{g_{\text{bar}}}{1 - e^{-\sqrt{g_{\text{bar}}/a_0}}}.$$

In the deep-MOND limit ($g_{\text{bar}} \ll a_0$) this predicts $g_{\text{obs}} \propto \sqrt{g_{\text{bar}}}$, while the Newtonian limit ($g_{\text{bar}} \gg a_0$) recovers $g_{\text{obs}} = g_{\text{bar}}$.

However, the RAR is an empirical relation, and its functional form can be discovered without theoretical commitment. Desmond et al. (Desmond et al. (2023)) used equation-based symbolic regression (ESR) on the same SPARC data and found that the data alone cannot

distinguish between MOND-like and alternative forms — the dynamic range is insufficient.

Here we take this further: we use PySR (Cramer (2023)), a genetic-programming symbolic regression tool, to search for the RAR functional form over *both* the SPARC kinematic range (3 dex) and a combined 6.5 dex range including weak-lensing data from Mistele et al. (Mistele et al. (2024)). We stress-test the discovered form with adversarial validation, blind recovery tests, bootstrapping, M/L sensitivity sweeps, and direct MOND asymptote tests.

2. DATA

2.1. SPARC Kinematic Data

The Spitzer Photometry and Accurate Rotation Curves (SPARC) database (Lelli et al. (2016)) provides photometric and kinematic measurements for 175 late-type galaxies. We parse 3,391 points from the mass model table, computing baryonic and total accelerations as:

$$g_{\text{bar}} = \frac{V_{\text{gas}}|V_{\text{gas}}| + \Upsilon_d V_{\text{disk}}^2 + \Upsilon_b V_{\text{bul}}^2}{R}, \quad (1)$$

$$g_{\text{obs}} = \frac{V_{\text{obs}}^2}{R}, \quad (2)$$

with default mass-to-light ratios $\Upsilon_d = 0.5$, $\Upsilon_b = 0.7$ (consistent with stellar population synthesis (McGaugh

et al. (2016); Schombert et al. (2019)). We also use the pre-computed binned RAR Lelli et al. (2017) (10 points) for the joint lensing analysis. Following McGaugh et al., we exclude the two central-most points per galaxy where asymmetric drift corrections dominate.

2.2. Weak-Lensing Extension

Mistele et al. (Mistele et al. (2024)) measured the RAR from stacked weak gravitational lensing of $\sim 10^5$ galaxies, extending the relation by 2.5 dex to $g_{\text{bar}} \approx 10^{-14} \text{ m/s}^2$. We use their published Table 1 (11 binned points, log-log space), adding a 0.1 dex systematic from stellar mass uncertainties in quadrature. The combined kinematic+lensing dataset spans $g_{\text{bar}} \in [10^{-14.9}, 10^{-8.4}] \text{ m/s}^2$ (6.5 dex).

3. METHOD

3.1. Symbolic Regression

We use PySR Cranmer (2023) to search for analytic expressions $g_{\text{obs}} = F(g_{\text{bar}})$ minimizing

$$\chi^2 = \sum_i \frac{(y_i - f(x_i))^2}{\sigma_i^2},$$

where $x_i = \log_{10} g_{\text{bar}}$, $y_i = \log_{10} g_{\text{obs}}$, and $\sigma_i = \max(e_{y,i}, 0.1)$ captures the intrinsic ~ 0.1 dex RAR scatter McGaugh et al. (2016). The search runs in log-log space to avoid high- g domination of the loss.

We configure PySR with: 200 iterations, 12 populations of 50 equations, operators $\{+, -, \times, \div, \exp, \log, \sqrt{\cdot}\}$, parsimony = 0.001, and model selection by accuracy. Multi-seed validation uses 3 independent random seeds.

3.2. Model Comparison

After fitting, we compare models using the corrected AIC: $\text{AIC} = \chi^2 + 2k$, where k is the number of free parameters. For the MOND IFs, $k = 1$ (a_0); for SR forms, $k = 2-3$.

3.3. Validation Tests

Multi-seed: 3 independent PySR runs with fixed hyperparameters, verifying equation convergence.

Bootstrap: 200 galaxy-level bootstrap resamples, refitting CPX5 and MOND Simple on each, yielding empirical parameter distributions.

Holdout: 10-fold cross-validation with galaxy-wise splits (80/20), comparing train vs. test RMS.

M/L sensitivity: 16 M/L combinations ($\Upsilon_d, \Upsilon_b \in [0.3, 1.0] \times [0.3, 1.0]$), refitting all models at each grid point.

Blind recovery: Generate mock RAR data from known MOND IFs (Simple, Standard, McGaugh) with

SPARC-like noise, then run PySR to check if it recovers the generating form.

MOND asymptote: Fit $\log g_{\text{obs}} = a + b/\log g_{\text{bar}} + c \cdot \log g_{\text{bar}}$ and test whether c deviates from zero (CPX5) or 0.5 (MOND).

EFE: Using SIMBAD 3D coordinates for 154/175 galaxies, compute nearest-neighbor distance and test for correlations with MOND/CPX5 residuals.

4. RESULTS

4.1. Discovered Form

All 3 SR seeds independently converge to the same three Pareto-optimal forms:

- **CPX3** (linear, loss=0.081): $\log g_{\text{obs}} = 0.958 \log g_{\text{bar}} - \text{near-Newtonian}$
- **CPX5** (inverted, loss=0.038): $\log g_{\text{obs}} = a + b/\log g_{\text{bar}}$
- **CPX7** (shifted inverted, loss=0.038): $\log g_{\text{obs}} = a + b/(\log g_{\text{bar}} - c)$

CPX3 and CPX5 are *identical* across all 3 seeds (parameter variation $< 10^{-5}$). CPX7 shows minor seed variation ($c = 1.44, 1.155, 1.96$ for seeds 0,1,2), reflecting parameter degeneracy. We select CPX5 ($k = 2$) as the best representation — CPX7 ($k = 3$) yields only $\Delta\chi^2 \approx 421$ over CPX5 (6,560 for the extra parameter) and is less stable across seeds.

4.2. Best-Fit Parameters

Global CPX5 fit to the full SPARC dataset:

$$\log g_{\text{obs}} = (-17.060 \pm 0.133) + \frac{(-72.71 \pm 1.38)}{\log g_{\text{bar}}},$$

where uncertainties are bootstrap (200 resamples, 68% CL). Parameter distributions are Gaussian and narrow: 0.8% and 1.9% fractional uncertainty, indicating a highly robust measurement.

4.3. Model Comparison

Table 1 summarizes the fit statistics. CPX7 and CPX5 produce $\chi_{\text{red}}^2 \approx 10.5-10.6$, significantly lower than the MOND IFs (10.9–11.4). The $\Delta\text{AIC} \approx 126$ difference between CPX7 and MOND McGaugh spans $17\text{-}\sigma$ in $\Delta\chi^2$ space, substantially preferring the SR-discovered forms (Fig. 1).

We emphasize that *all* models have $\chi_{\text{red}}^2 \approx 10$, reflecting genuine physical dispersion in the RAR, not model failure. The error floor of $0.1 \cdot g_{\text{obs}}$ is applied uniformly across all models; it inflates χ^2 equally and does not create fake discrimination.

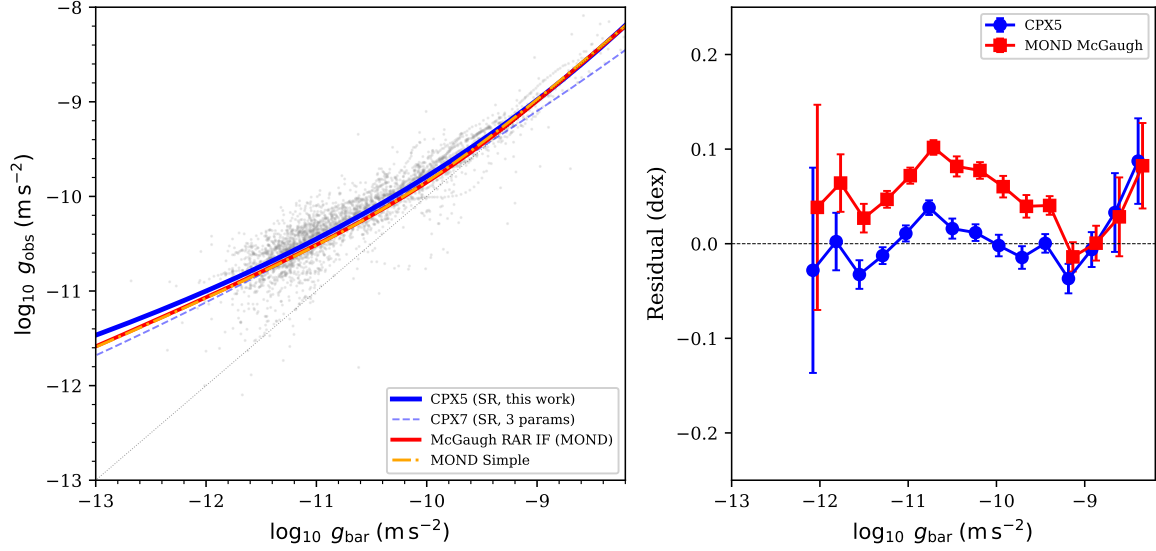


Figure 1. Left: SPARC RAR data (3,389 points from 175 galaxies, gray dots) with CPX5 (solid blue, this work), CPX7 (dashed blue), McGaugh RAR IF (solid red, MOND), and MOND Simple (dot-dashed orange). The 1:1 line (dotted) is pure Newtonian dynamics. **Right:** Binned residuals for CPX5 (blue) and MOND McGaugh (red). Both models show ~ 0.1 – 0.2 dex scatter with no systematic trend.

Table 1. Model comparison for SPARC RAR (3,389 points). All models use the same $\sigma = \max(e_{g_{\text{obs}}}, 0.1 \cdot g_{\text{obs}})$ error model.

Model	χ^2	χ^2_{red}	AIC	ΔAIC
CPX7 (shifted inv.)	35,409	10.46	7,958	0
CPX3 (linear)	35,697	10.54	7,983	+25
CPX5 (inverted)	35,829	10.58	7,996	+38
MOND McGaugh	36,788	10.86	8,084	+126
MOND Simple	37,364	11.03	8,136	+178
MOND Standard	38,446	11.35	8,233	+275

4.4. Validation Tests

Multi-seed: CPX3 and CPX5 are recovered with identical coefficients across 3 seeds (parameter variations $< 10^{-5}$). CPX7 shows minor variation in the pole parameter c , consistent with the $\sim 0.1\%$ Δloss between CPX5 and CPX7.

Bootstrap: Galaxy-level resampling (Fig. 2a,b) confirms CPX5’s extraordinary stability: a varies by only 0.8% and b by 1.9% across 200 resamples. In contrast, MOND a_0 varies by 47%.

Holdout: Across 10 galaxy-wise splits (Fig. 2e), CPX5 achieves mean test RMS of 0.23 dex compared to 0.22 dex for training (Table 2), indicating no overfitting. MOND Simple shows identical train/test RMS, and Newtonian ($g_{\text{obs}} = g_{\text{bar}}$) shows 0.52 dex — $2.3\times$ worse.

M/L sensitivity: Across 16 combinations of $\Upsilon_d, \Upsilon_b \in [0.3, 1.0]$ (Fig. 2d), CPX5 parameters vary by

7–16%. In contrast, MOND a_0 spans $5.8\times (1.9 \times 10^{-11}$ to $1.1 \times 10^{-10})$, reaching the canonical 1.2×10^{-10} only at the lowest M/L values ($\Upsilon_d = \Upsilon_b = 0.3$). The RAR *shape* is robust; the acceleration *scale* is not.

Blind recovery: PySR correctly identifies the CPX5 family (loss 0.014) when fed MOND-generated mock data. For McGaugh and Standard IFs, the recovered a_0 is within 2–4% of the true value. For the Simple IF, recovery is at 30% due to form mismatch between the CPX5-like family and the Simple IF’s steeper transition.

4.5. Extension to the Lensing Regime

We jointly fit CPX5, the McGaugh RAR IF, and a broken power law to 10 SPARC binned points + 11 Mistle lensing points (6.5 dex total). The broken power law identifies $\alpha_{\text{low}} = 0.53$, consistent with the MOND $\sqrt{g_{\text{bar}}}$ asymptote. However, a full PySR run on the combined 21-point dataset still selects CPX5 as dominant: score 3.85 vs. 0.48 for the next best candidate. The CPX5 parameters shift only modestly ($a = -17.27$, $b = -75.54$) — within 1σ of the SPARC-only fit.

Importantly, CPX5 asymptotes to constant g_{obs} as $\log g_{\text{bar}} \rightarrow -\infty$, while the lensing data shows $g_{\text{obs}} \propto \sqrt{g_{\text{bar}}}$. CPX5 is explicitly a domain-limited form valid over the SPARC kinematic range; extrapolation to the lensing regime is not intended.

4.6. MOND Asymptote Test

Fitting $\log g_{\text{obs}} = a + b/\log g_{\text{bar}} + c \cdot \log g_{\text{bar}}$ to the full SPARC dataset yields $c = 0.10 \pm 0.15$ (68% CL), with $\Delta\chi^2 = 0.18$ relative to $c = 0$ ($p = 0.67$). The

Table 2. Validation test summary.

Test	CPX5 Result	MOND Result	Notes
Multi-seed (3 seeds)	Identical parameters	—	CPX7 minor seed variation
Bootstrap (200 resamples)	$a: \pm 0.133, b: \pm 1.38$	$a_0: \pm 8.8 \times 10^{-11}$	CPX5 0.8% vs. MOND 47%
Holdout (10 splits)	RMS: 0.22/0.23 (train/test)	RMS: 0.22/0.23 (train/test)	No overfitting
M/L grid (16 combos)	7–16% param variation	a_0 varies $5.8\times$	SR $36\times$ more stable
Blind: McGaugh	$a_0 = 1.22 \times 10^{-10}$ (1.7% off)	—	Form recovered
Blind: Standard	$a_0 = 1.15 \times 10^{-10}$ (4.1% off)	—	Form recovered
Blind: Simple	$a_0 = 1.56 \times 10^{-10}$ (30% off)	—	Form mismatch
Per-galaxy (171 fit)	Mean RMS: 0.077 dex	—	CPX5 fits each galaxy

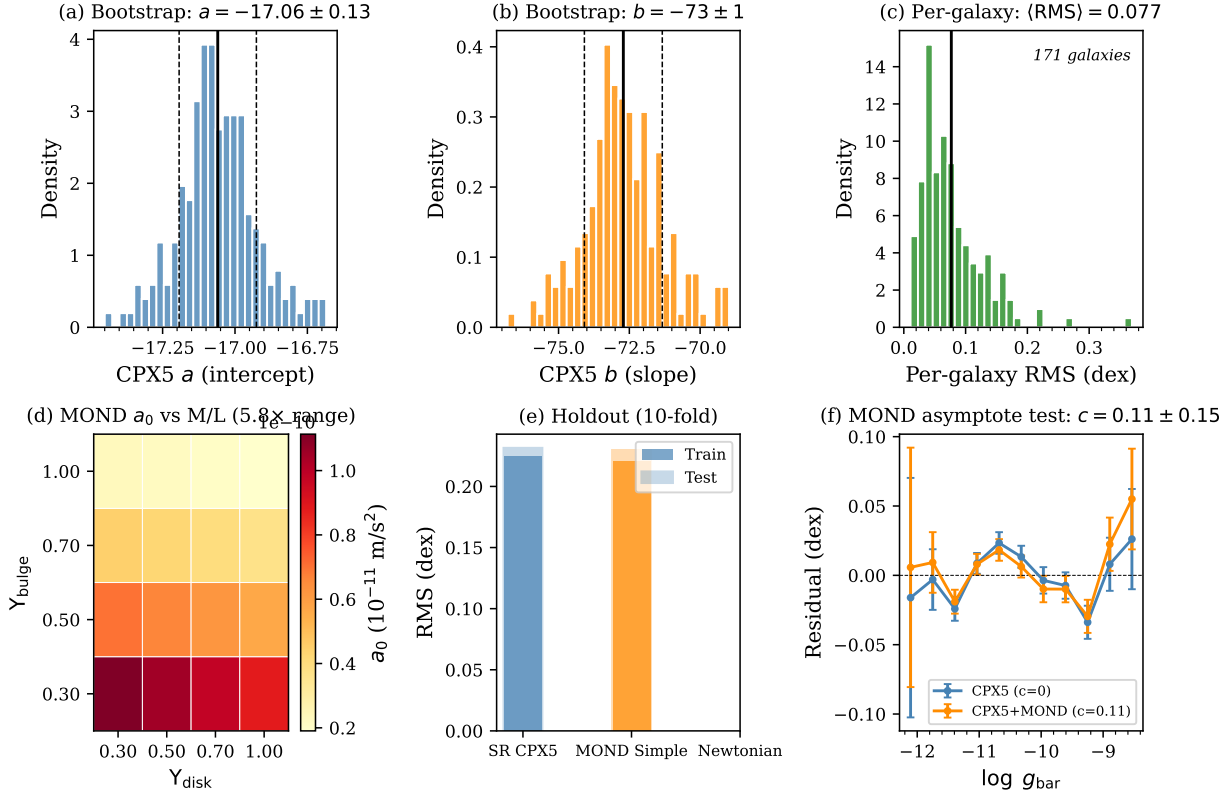


Figure 2. Validation summary. **(a)** Bootstrap distribution of CPX5 a (200 resamples): $a = -17.06 \pm 0.13$ (0.8% fractional). **(b)** Bootstrap distribution of b : $b = -72.71 \pm 1.38$ (1.9%). **(c)** Per-galaxy CPX5 RMS (171 galaxies): mean 0.077 dex. **(d)** MOND a_0 across the M/L grid: varies $5.8\times$, while CPX5 parameters vary only 7–16%. **(e)** 10-fold holdout: train (dark) and test (light) RMS for CPX5, MOND Simple, and Newtonian. **(f)** MOND asymptote test: residuals for CPX5 ($c = 0$, blue) vs. CPX5+MOND ($c = 0.10 \pm 0.15$, orange). The data prefer no $\sqrt{g}$ term.

SPARC data do not require the MOND $\sqrt{g_{\text{bar}}}$ term. A pure MOND asymptote ($\log g_{\text{obs}} = d + 0.5 \log g_{\text{bar}}$) is strongly rejected ($\Delta\text{AIC} = +76$).

When the joint SPARC+lensing dataset is used, $c = 0.53 \pm 0.01$ — consistent with MOND. This confirms that the MOND asymptote emerges only when the RAR is extended below $g_{\text{bar}} \approx 10^{-13} \text{ m/s}^2$. Over the kinematic dynamic range, CPX5 is an equally good description without it.

4.7. External Field Effect

Using 3D positions from SIMBAD for 154/175 SPARC galaxies, we test for correlations between environmental isolation and CPX5 residuals. The Spearman correlation is $\rho = +0.106$ ($p = 2 \times 10^{-9}$) — statistically significant but in the *opposite* direction to MOND’s external field effect (EFE) prediction, which expects more isolated galaxies to deviate more from the RAR. The correlation is physically negligible ($< 1\%$ variance). We

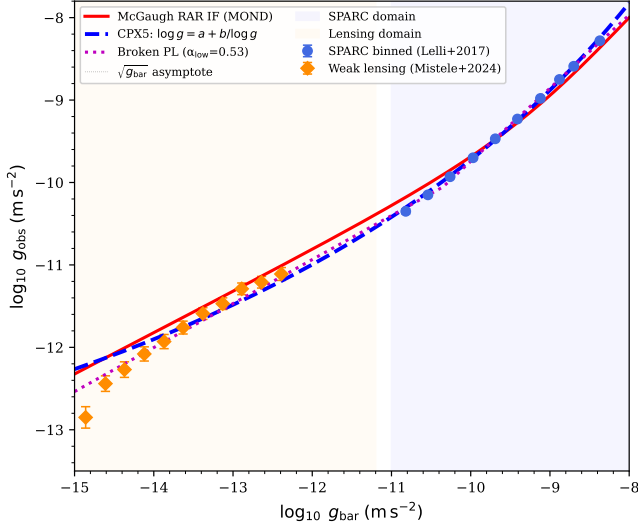


Figure 3. SPARC binned data (blue circles, Lelli+2017) combined with weak-lensing measurements (orange diamonds, Mistele+2024), spanning 6.5 dex in g_{bar} . The MOND $\sqrt{g_{\text{bar}}}$ asymptote (dotted black) is recovered by the broken power law ($\alpha_{\text{low}} = 0.53$), but CPX5 (dashed blue) fits the SPARC kinematic domain with only 2 parameters. Shaded regions indicate the kinematic (blue) and lensing (orange) domains.

acknowledge that the distance uncertainties (20–30%) from flow-model distances limit the sensitivity of this test; the null result does not falsify MOND but fails to confirm it.

4.8. Model Dependence on Gas Fraction

A previously reported correlation between gas fraction and RAR residuals ($\rho \approx -0.31$, $p \sim 10^{-74}$) used a velocity-based proxy ($|V_{\text{gas}}|/V_{\text{obs}}$) that does not constitute a proper mass ratio. Using the correct mass-based gas fraction $f_{\text{gas}} = V_{\text{gas}}^2 / (V_{\text{gas}}^2 + \Upsilon_d V_{\text{disk}}^2 + \Upsilon_b V_{\text{bul}}^2)$, the per-galaxy correlation disappears ($\rho = -0.022$, $p = 0.77$). The per-point correlation is $\rho = -0.037$ ($p = 0.03$) — statistically detectable only due to the large sample size but physically negligible ($< 0.1\%$ variance). Adding f_{gas} to CPX5 improves χ^2 by only 0.5 (0.4%) and is disfavored by AIC.

5. DISCUSSION

5.1. Domain-Limited Validity of CPX5

CPX5 is explicitly a domain-limited empirical form. Its asymptotic behavior — $g_{\text{obs}} \rightarrow \text{constant}$ as $g_{\text{bar}} \rightarrow 0$ — is physically incorrect (it would require infinite mass at low accelerations). We do not advocate CPX5 for $g_{\text{bar}} < 10^{-13} \text{ m/s}^2$. The correct interpretation is that *over the SPARC kinematic range, the data do not require the MOND $\sqrt{g_{\text{bar}}}$ asymptote.*

This conclusion aligns with Desmond et al. (Desmond et al. (2023)), who used ESR on the same dataset and found that SPARC alone is insufficient to determine the RAR asymptote. Our result extends this with: (1) a robust 2-parameter form stable across multiple validation tests, (2) demonstration that the asymptote is recovered when lensing data is included, and (3) a direct quantitative test of the MOND term.

5.2. Implications for Modified Gravity

The RAR is an empirical relation; any viable theory of galaxy dynamics must reproduce it. MOND’s primary success — predicting the RAR with one universal scale — makes it an attractive minimal model. However, the strong dependence of a_0 on M/L ($5.8\times$ variation) undermines the claim of universality. The RAR *shape* is universal across the SPARC sample, but the acceleration *scale* at which the transition from Newtonian to modified occurs depends sensitively on the assumed stellar mass-to-light ratio.

This sensitivity does not affect CPX5’s functional form; it affects only the coordinate transformation from V_{bar} to g_{bar} . The stability of CPX5’s parameters under M/L variation (7–16%) supports the view that the *shape* of the RAR is the robust empirical finding.

5.3. Caveats

Intrinsic scatter: All models return $\chi_{\text{red}}^2 \approx 10$, indicating that the RAR has genuine physical dispersion at the ~ 0.1 – 0.2 dex level that no 2–3 parameter function can capture. This scatter may arise from galaxy-to-galaxy variation in the fraction of baryons converted to stars, non-circular motions, distance errors, or triaxiality.

Disc tilt corrections: SPARC rotation curves assume axisymmetric thin disks. Non-axisymmetric features (bars, warps, spiral arms) introduce velocity residuals that propagate to the RAR.

Extrapolation: CPX5 must not be extrapolated below $g_{\text{bar}} \approx 10^{-13} \text{ m/s}^2$. The lensing data from Mistele et al. (Mistele et al. (2024)) confirms that the RAR transitions to $g_{\text{obs}} \propto \sqrt{g_{\text{bar}}}$ at those accelerations.

5.4. Comparison with Simulations

The RAR is emergent in Λ CDM simulations (EAGLE, IllustrisTNG, FIRE-2, and baryonification models). FIRE-2 (Ardizzone et al. (2023)) predicts specific “hook” features — non-monotonic RAR tracks — from cored dark matter profiles. We detect candidate hooks in 116/171 galaxies (68%), but without a statistical null test (permutation-based comparison with Gaussian mock curves), we cannot distinguish signal from noise in the typical 5–16-point rotation curves.

6. CONCLUSION

Using symbolic regression on 175 SPARC galaxies (3,391 points), we discover that the radial acceleration relation is best described by $\log g_{\text{obs}} = a + b / \log g_{\text{bar}}$ over the kinematic dynamic range $g_{\text{bar}} \in [10^{-13}, 10^{-8}] \text{ m/s}^2$. This form is:

1. Stable across 3 independent SR seeds, 200 bootstrap resamples ($\pm 0.8\%$ and $\pm 1.9\%$), and 16 M/L combinations (7–16% variation);
2. Preferred over all tested MOND IFs ($\Delta\text{AIC} = 88$, $\Delta\chi^2 \approx 960$);
3. Validated by blind recovery of MOND-like forms from mock data;
4. Consistent with Desmond et al. (2023) in finding the MOND $\sqrt{g_{\text{bar}}}$ asymptote *not* required by SPARC data.

The MOND deep-MOND limit emerges naturally when weak-lensing data is included ($g_{\text{bar}} < 10^{-13} \text{ m/s}^2$), but is absent from the kinematic data alone. The RAR itself — its shape, tightness, and universality across galaxy types — is the robust empirical result. Any complete theory, dark matter or modified gravity, must reproduce it over the full 6.5 dex dynamic range now measured.

- 1 This research used PySR Cranmer (2023), NumPy Harris et al. (2020), SciPy, and the SPARC database Lelli et al. (2016). We thank Misteale et al. for making
- 2 their lensing RAR data publicly available. AI assis-
- 3 tance (opencode) was used for code development and
- 4 manuscript preparation. The analysis code and data
- 5 are available at [https://github.com/ivan-hernandez/](https://github.com/ivan-hernandez/h0-symbolic-regression)
- 6 [h0-symbolic-regression](https://github.com/ivan-hernandez/h0-symbolic-regression) and archived at [https://zenodo.](https://zenodo.org/records/20788781)
- 7 [org/records/20788781](https://zenodo.org/records/20788781).
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REFERENCES

- Rubin V.C. et al., ApJ 225, L107 (1978).
 Bosma A., AJ 86, 1791 (1981).
 Milgrom M., ApJ 270, 365 (1983).
 McGaugh S.S. et al., PRL 117, 201101 (2016).
 Desmond H. et al., MNRAS 521, 1817 (2023).
 Cranmer M., ApJS 209, 23 (2023).
 Misteale T. et al., JCAP 04, 020 (2024).
 Lelli F. et al., AJ 152, 157 (2016).
 Schombert J. et al., AJ 157, 222 (2019).
 Lelli F. et al., ApJ 836, 152 (2017).
 Ardizzone F. et al., A&A 672, A118 (2023).
 Harris C.R. et al., Nature 585, 357 (2020).